

Prateek Tandon

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SUMMARY

Results-driven Data Engineer and Data Scientist with 10+ years of experience designing and scaling robust ETL pipelines and workflows on AWS delivering robust, high-quality code within the SDLC framework, including on ISO 13485-certified projects. Proven expertise in machine learning and statistics, with a strong focus on data quality, compliance, and operational efficiency. Adept at architecting production-grade systems to process high-volume datasets across hybrid cloud infrastructures, while collaborating closely with analytics, product, and ML teams to drive business impact.

SKILLS

Programming	Python Bash Java SQL
Database	Snowflake MSSQL Server Oracle 10g MySQL PostgreSQL Sqlite
Python toolkit	pyspark pandas multiprocessing seaborn flask sqlalchemy mypy streamlit jinja2
Python ML	tensorflow scikit-learn pytorch statsmodels openAI nltk OpenCV XGBoost
Statistics	PCA KS-test KL divergence K-means Student's t-test Mann-Whitney U test
Tools	Git Subversion JIRA Agile Ant Maven Jupyter VSCode GitHub Actions
Cloud Stack	AWS (IAM EC2 S3 RDS EFS DynamoDB ELB API Gateway Lambda SQS SNS)
Infrastructure	Terraform Ansible Docker Jenkins Travis CI Prometheus Grafana
HPC & ETL/ELT	Sun Grid Engine (SGE) Ganglia SJM Flower Snakemake Apache Airflow
Familiarity	XML JSON R JavaScript D3.js HTML C++ MATLAB RAGs Databricks

EXPERIENCE

Senior Data Scientist & Data Engineer	Illumina Inc.	Nov 2022 – Present
- Led development of DVTRunner tool (Docker+HPC+Snakemake+Jenkins), enabling scientists to launch & monitor clinical workflows - Applied CI/CD best practices during dev lifecycle ensuring reliable feature delivery, high system availability & smooth user workflows - Advised peers on effective DVTRunner & HPC cluster usage & troubleshooting techniques; achieved zero unresolved issues. - Managed a 25 server FPGA cluster, improving throughput & minimizing downtime for high-throughput clinical ETL pipelines - o Improved system observability and automation for alerting features with Prometheus and Grafana - o Implemented Isilon data lifecycle monitoring with AWS S3 and S3 Deep Glacier Archive - o Refactored Ansible playbooks to ensure idempotent server provisioning, reducing configuration drifts and errors - o Maintained FPGA cluster task database and migrated schema using Alembic - o Led server OS migration project (CentOS7 -> Oracle Linux 8) performing comprehensive equivalency studies - Led analytics reporting service development (powered by Snowflake, REST APIs & Jinja2) delivering scheduled and on-demand insights into project progress, failure trends etc. for C-suite executives & senior stakeholders' consumption - Designed the Snowflake DB schema for storing sample analysis metrics along with unstructured OMOP clinical metadata - Led perturbation analysis supporting stability studies during assay development with background distribution estimation - Applied generative modeling to simulate variants in expanded genomic regions, evaluated PPA & PPV to guide product design decisions		
Technical Lead Bioinformatics Engineering	Personalis Inc.	July 2018 – Nov 2022
- Led analytical validation of SNVs, Indels, Gene Fusions, CNVs, TMB, MSI, TCR-β biomarker discovery and tumor profiling Dx products - Analysed early-stage R&D feasibility data for NeXT Personal[®] liquid biopsy product, shaping core features and workflows. - Built Python API for accessing pipeline metadata & QC metrics, enabling rapid triage & standardized downstream workflows - Served as on-call data engineer, supporting production analyses, ensuring rapid resolution & minimally disrupted critical workflows - Led DNA small variants and RNA gene fusions detection algorithms evaluation project & implemented data-driven random forests-based filtering steps to accelerate runtime and improve F1-score for WES and WTS data - Led the consolidation of verification and validation analysis workflows into a unified, version-controlled and packaged codebase, improving maintainability, collaboration, and reproducibility across teams		
Bioinformatics NLP Engineer	QIAGEN Inc.	Aug. 2014 – July 2018
- Leveraged NLP models (Word2Vec, LSTM, LDA) to ensure scientific accuracy of ontology-driven insights from biomedical text data - Implemented ontology powered NLP based tests to disease-to-drug association content quality in the knowledge base - Implemented python API for the knowledge base powering Ingenuity Pathway Analysis (IPA) and QIAGEN Clinical Insight (QCI) - Extended in-house Java CI framework with features and reports to ensure complete testing of new functionalities		

- Acted as Scrum Master and facilitated requirements gathering and feature prioritization from product managers and between domain scientists and engineering, accelerating delivery of clinical-grade variant classification tools in QCI
- Developed CI test suites for 3rd party data integration (RefSeq, Ensembl, HGMD, Entrez Gene, ClinicalTrials, FDA, gnomAD, ClinVar)
- Implemented Flask+SQLite based web applications for existing script-based tools to improve curation accuracy and efficiency

SIDE-PROJECTS

ETL infrastructure provisioning on AWS via Terraform (https://github.com/erprateek/backend_etl-infra)

- Infrastructure-as-Code (IaC) repository for provisioning and managing backend infrastructure on AWS (DynamoDB, S3, Lambda function that listens to new files in S3 bucket and triggers ETL pipeline), using Terraform and supporting shell scripts.
- Successfully resolved encryption vulnerabilities reported by tfsec security scanner GitHub CI/CD workflow

Monte Carlo Simulation of Somatic SNVs (https://github.com/erprateek/somatic_snv_simulator)

- Implemented dynamic memory partitioning at runtime while generating Poisson & Binomially distributed variables for 3B simulations
- Optimized to run 3-billion simulations in a reasonable span of wall clock time (12 mins) on laptop with 4Gb physical memory

LLM Pipeline Explainer (<https://github.com/erprateek/pipeline-explainer>)

- Developed an LLM-driven streamlit web application using Mixtral-8x7B-Instruct-v0.1 to parse and summarize Snakemake, Nextflow or Shell script bioinformatics workflows, aiding debugging and knowledge transfer for multidisciplinary teams

Containerized HPC environment emulation (<https://github.com/erprateek/hpc-celery-cluster>)

- Implemented a POC distributed task queue system utilizing FastAPI as the master application, Celery for task management, and Redis as the message broker.

EDUCATION

Carnegie Mellon University	M.S. Computational Biology	May 2014
Vellore Institute of Technology	B. Tech. Bioinformatics	May 2012

Relevant Coursework: Principles of software construction | Machine learning | String algorithms | Algorithms & data structures | Web application development | Database Management Systems | Probability and statistics

PROFESSIONAL DEVELOPMENT

Docker Foundations Prof. Cert.	LinkedIn Learning	May 2025
Data Pipeline Management with Apache Airflow	LinkedIn Learning	May 2025
Leadership Foundations	LinkedIn Learning	May 2025
AWS Cloud Solutions Architect, Prof. Cert.	AWS Coursera	Jan 2024
AWS Cloud Technology Consultant Prof. Cert.	AWS Coursera	Jan 2024
Machine Learning	DeepLearning.AI Coursera	Jul 2023
Architecting on AWS	Amazon Web Services	Feb 2022
DeepLearning.AI TensorFlow Developer	DeepLearning.AI Coursera	Oct 2020

PUBLICATIONS

- **[POSTER] Analytical performance of the new Illumina TruSight™ Oncology 500 v2 assay for comprehensive genomic profiling (AMP 2024)**
Danny M. Chou, Jennifer L. Silhavy, Brian Crain, Austin Purdy, Kaitlin O'Brien, Urminder Singh, Sosse Agvanian, Nancy Valencia, Jacqueline Ling, Jonathan Rodriguez, Blair Conner, **Prateek Tandon**, Roger C. Tim, Jennifer S. LoCoco, Scott Lang, Traci Pawlowski
- **[PATENT] Machine-learning techniques for predicting surface-presenting peptides (18/065,410)**
Charles Wilbur ABBOTT III, Sean Michael Boyle, Rachel Marty Pyke, Eric Levy, Dattatreya Mellacheruvu, Rena McClory, Richard Chen, Robert Power, Gabor Bartha, Jason Harris, Pamela Milani, **Prateek Tandon**, Paul McNitt, Massimo Morra, Sejal Desai, Juan-sebastian Salvidar, Michael Clark, Christian Haudenschild, John West, Nick Phillips, Simo V Zhang
- **[PATENT] Composite biomarkers for immunotherapy for cancer (17/965,719)**
Charles Wilbur ABBOTT III, Sean Michael Boyle, Rachel Marty Pyke, Eric Levy, Dattatreya Mellacheruvu, Rena McClory, Richard Chen, Robert Power, Gabor Bartha, Jason Harris, Pamela Milani, **Prateek Tandon**, Paul McNitt, Massimo Morra, Sejal Desai, Juan-sebastian Salvidar, Michael Clark, Christian Haudenschild, John West, Nick Phillips, Simo V Zhang
- **[PAPER] Analytic Validation of NeXT Dx™, a comprehensive genomic profiling assay – (Oncotarget 2023)**
Saldivar J., Harris J., Ayash E., Hong M., **Tandon P.**, Sinha S., Hebron P. Miranda, Houghton E. E., Thorne K., Goodman L. J., Li C., Marfatia T. R., Anderson J., et al. Oncotarget, 2023; 14: 789-806. DOI: 10.18632/oncotarget.28490